

Full Length Article

Dynamic-Aware video distillation: Adaptive temporal partitioning based on video semantics for edge device

Yinjie Zhao^a, Heng Zhao^b, Yew-Soon Ong^b, Bihan Wen^a, Joey Tianyi Zhou^{b,*}^a Nanyang Technological University, Singapore^b Agency for Science Technology and Research, Singapore

ARTICLE INFO

Keywords:

Dataset distillation

AI Efficiency

ABSTRACT

Deep learning model training on edge device is challenged by limited computational power and dataset scale. Dataset distillation (DD) aims to address this challenge by synthesizing a compact, small-scale dataset that yields model performance comparable to the full original data. While DD has achieved notable success on image datasets, existing approaches struggle with video data due to its inherent temporal complexity and varying redundancy across different video semantics. Prior methods assume uniform level of temporal redundancy across all video semantics, which limits their effectiveness. In this work, we propose Adaptive Temporal Partitioning of synthetic videos to account for semantic adaptability and better temporal redundancy reduction. We further propose Dynamic-Aware Video Distillation (DAViD), a reinforcement learning framework that learns an optimal temporal partitioning policy. A teacher-in-the-loop reward function guides the RL agent's policy updates. To our knowledge, this is the first study to adaptively reduce temporal redundancy based on video semantics within the DD domain. Our approach substantially outperforms existing DD techniques, demonstrating significant gains in efficiency and model performance. We believe this work paves way for more efficient, semantics-aware video DD for resource-limited edge device scenarios.

1. Introduction

The development of edge devices has led to an ever-increasing demand for efficient video understanding on resource-constrained platforms, and on-device training techniques are of increasing importance for edge AI systems. Under low communication budget scenarios, it is infeasible to train deep learning model off-device and upload the model to the device due to either high latency or low bandwidth Lopez-Paz and Ranzato (2019), Singh and Gill (2023). Facing personalization and privacy reserving requirements, training data is also expected to remain on-device, and therefore on-device models need to be trained locally Jouini et al. (2024), KHAN et al. (2016), Liu et al. (2025), Wang et al. (2025). Furthermore, on-device models are expected to adapt to newly emerging and unique data distributions, where off-device training are largely challenged by generalizability issues Wang et al. (2024a).

Unlike conventional hardware for AI training, edge devices are limited in compute power, memory capacity, and energy budgets. For instance, NVIDIA Jetson Orin Nano operates under maximum 8 GB RAM¹,

and Raspberry Pi 5 under maximum 16GB RAM². Therefore the deployment of large-scale video recognition models or training on massive datasets is infeasible due to such computational constraints. Under such constraints, on-device learning requires both lightweight architectures and compact datasets. One promising direction is Dataset Distillation (DD) Wang et al. (2018), which synthesizes small yet informative datasets that preserve the representational power of the original data while drastically reducing storage and computation costs. These efficiency gains are crucial for enabling scalable and practical video intelligence on device.

Existing video Dataset Distillation (DD) approaches often compress an entire synthetic video into a monolithic sequence Wang et al. (2024b). Specifically, existing approaches largely ignored that different parts of a video carry different semantic meaning and temporal

² See description from <https://www.raspberrypi.com/products/raspberry-pi-5/>

* Corresponding author.

E-mail addresses: yzhao0267@gmail.com (Y. Zhao), zhao_heng@cfar.a-star.edu.sg (H. Zhao), ASYSOng@ntu.edu.sg (Y.-S. Ong), bihan.wen@ntu.edu.sg (B. Wen), joey_zhou@a-star.edu.sg (J.T. Zhou).

¹ See description from <https://www.nvidia.com/en-sg/autonomous-machines/embedded-systems/jetson-orin/nano-super-developer-kit>

<https://doi.org/10.1016/j.neunet.2026.108871>

Received 4 September 2025; Received in revised form 13 March 2026; Accepted 16 March 2026

Available online 29 March 2026

0893-6080/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

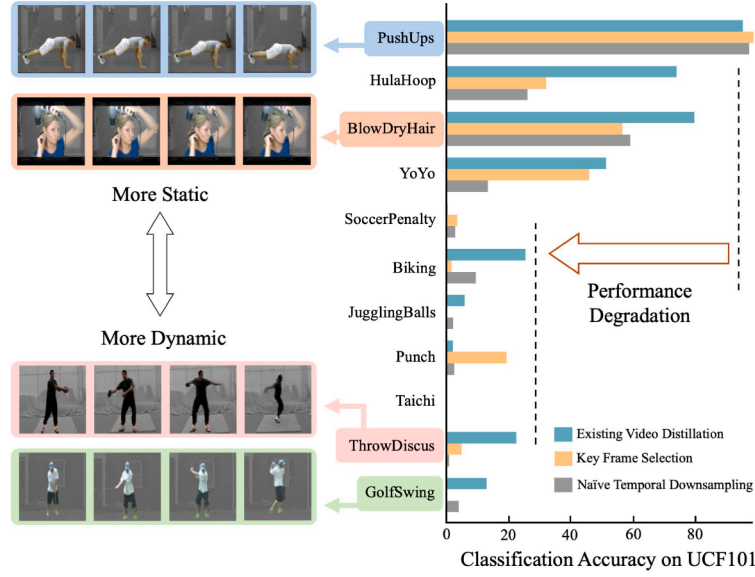


Fig. 1. Performance degradation on more dynamic video classes when trained on distilled dataset. When constructing a compact video dataset, conventional temporal downsampling and existing video DD methods Wang et al. (2024b) often suffer from significant performance degradation, particularly in more dynamic video classes. This limitation arises because these approaches assumed monolithic temporal dimension for synthetic videos, and failed to adapt to different video semantics.

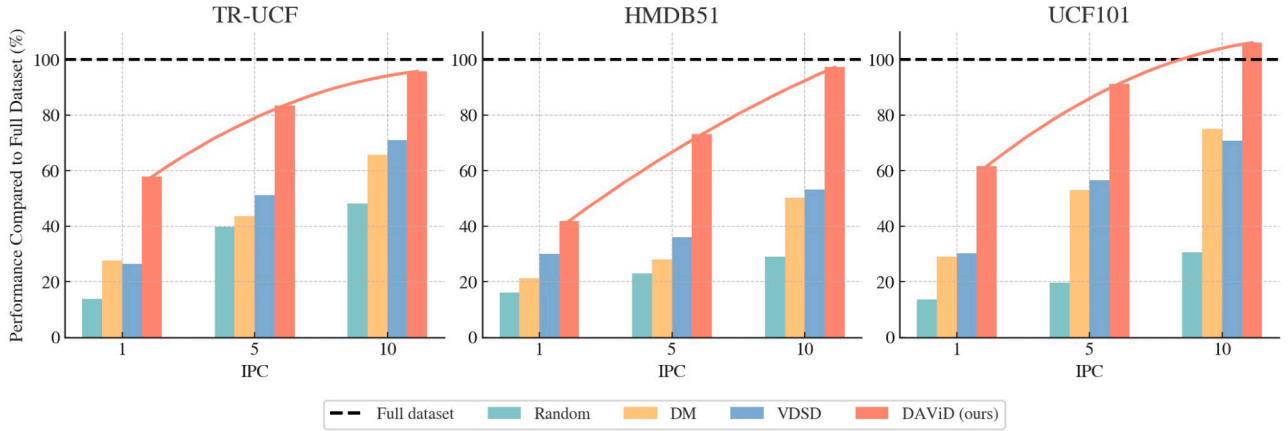


Fig. 2. Our approach obtained nearly lossless performance compared to the original dataset at IPC = 10, while existing approaches such as DM Zhao and Bilen (2023) and VDS Wang et al. (2024b) has shown a large performance gap from lossless dataset distillation. However, previous work VDS only showed similar performance with image level DD method (DM).

dynamics. This oversimplification leads to significant information loss, particularly for dynamic video classes where temporal cues are critical. The consequence is magnified in low memory scenarios, where there are low upper bounds for dataset sizes and performance degradation directly translates to poor deployment feasibility.

We can observe in Fig. 1, previous approaches to produce compact datasets are not able to faithfully represent complex temporal patterns. When training models on the distilled datasets, existing approaches suffered from severe performance degradation on more dynamic and complex video semantics. For example, actions such as *GolfSwing* usually requires multiple frames to be recognized, while *PushUps* is typically recognizable given one single frame. The gap of model performance between these semantics are significant, indicating the importance of adaptively utilize computational budget according to the temporal complexity. Furthermore, we observed an average of 70.0% performance loss compared to the original dataset for traditional optical-flow based key-frame selection across different datasets (see details in Table 1), and existing SOTA video DD approaches also produces an average 35.0% performance loss compared to the origi-

nal dataset. This demonstrated the large performance gap for existing research.

In contrast, human cognition process does not experience huge performance loss across different video dynamics. With constrained energy and effort, the brain is considered rather memory-efficient, storing dynamic visual information without compromising complex semantics. As illustrated by Fig. 3 (a), cognitive neuroscience suggests that the brain segments continuous streams of visual input into discrete “scene chunks” or events Ben-Yakov and Henson (2018), Kurby and Zacks (2008), Radvansky and Zacks (2014), Sargent et al. (2013), Zacks et al. (2001, 2007), storing these as cohesive memory units rather than isolated frames.

Inspired by neurocognitive “chunking”, we propose Adaptive Temporal Partitioning according to different video semantics. It aims to reduce the temporal redundancy for synthetic video datasets according to the demand of different video semantics. Specifically, a synthetic video is adaptively partitioned into variable-length segments along temporal dimension, each capturing a semantically coherent sub-event. Intuitively, segments where more complex scene contents such as faster

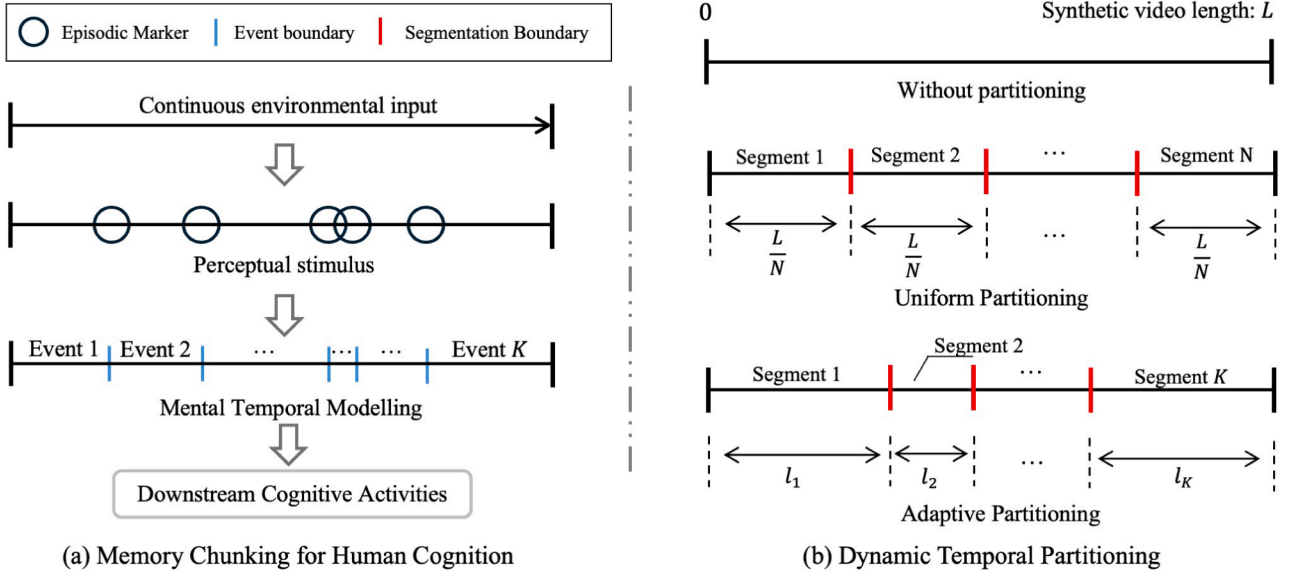


Fig. 3. (a) Chunking largely reduces temporal redundancies for human cognition to process dynamic visual inputs. Episodic markers are caused by salient perceptual stimulus, enabling a reconstruction of memories based on the markers and natural temporal partitioning. In this process, different temporal segments carry different information [Khemlani et al. \(2015\)](#). (b) Inspired by this, we propose to adaptively partition the synthetic video into multiple segments along temporal dimension. The distilled videos adapt to dynamic semantics with different complexities. First line: Existing approaches on video DD, Second line: Simple baseline of uniform partitioning, Third line: Adaptive partitioning proposed by us, where the total number and lengths of segments are optimized based on different video semantics.

motions or object interactions demand finer partitioning granularity (longer segments) to preserve detail, whereas more static intervals can be represented by shorter temporal spans.

Different levels of adaptiveness for temporal partitioning are illustrated in [Fig. 3 \(b\)](#). Without partitioning or uniform partitioning as intuitive baselines, is unable to adapt to diverse levels of semantical complexities. Our method aims to partition the synthetic video adaptively. This adaptive segmentation poses two intertwined challenges: (i) the number of segments per video is unknown and must be inferred, and (ii) determining each segment's boundaries is a discrete, non-differentiable decision, precluding direct gradient-based optimization.

To tackle these challenges in an computational efficient way, we cast the partitioning process as a sequential decision process and leverage Reinforcement Learning (RL) to learn adaptive partitioning policies. At each step, an RL agent observes the current temporal location and decides where to place the next temporal cut, thereby determining the length of the upcoming segment. Upon partitioning the entire synthetic video, the agent receives a reward computed as the feature-distance between models trained on the distilled set versus those trained on the original data. This sparse, delayed reward reflects the quality of the current partitioning strategy preserving the overall distribution of video semantics.

The state of the RL agent is the current temporal location; the action is length of next segment; and the reward function quantifies the distillation fidelity via feature-space distances. With this formulation, our RL-based DD pipeline automatically adapts both the number and lengths of segments in each synthetic video. Empirically, we demonstrate that our method yields higher downstream task accuracy compared to state-of-the-art dataset distillation baselines. Our main contributions can be summarized as the following:

- We propose Adaptive Temporal Partitioning for video DD, effectively adapting temporal granularity according to the complexity video semantics under resource-limited scenario.
- We propose an RL framework, specifically, a Q-learning process tailored for Dataset Distillation process with a Teacher-Guided Reward Function to solve such a partitioning problem.

- A SOTA performance with a large margin of improvement has been achieved on various video recognition benchmarks under low training resource scenarios.

2. Related work

2.1. Resource-limited video recognition training

Existing approaches for on-device training efficiency have been focusing on reducing memory and FLOPs consumptions with the original dataset unchanged [Cai et al. \(2020\)](#), [Gim and Ko \(2022\)](#), [Huang et al. \(2023\)](#), [Lee et al. \(2022\)](#), [Lin et al. \(2022\)](#), [Lopez-Paz and Ranzato \(2017\)](#), [Patil et al. \(2022\)](#), [Zhang et al. \(2020\)](#). Other approaches improve training efficiency by proposing efficient training techniques or model architectures [Kim et al. \(2023\)](#), [Lin et al. \(2020\)](#), [Nguyen et al. \(2025\)](#), [Paissan et al. \(2024\)](#), [Yang et al. \(2022\)](#). These approaches reduces training computational workload, but are unable to conduct dataset redundancy reduction, and therefore leave the data-efficiency challenge remain unaddressed.

Dataset Distillation has emerged in recent years to obtain a representative and compact dataset trained on limited resources, jointly addressing the problems of training efficiency and expensive dataset scaling. However, previous image-level Dataset Distillation (DD) [Wang et al. \(2018\)](#) approaches assumes monolithic outputs. The synthetic datasets are usually obtained as a whole by matching the features of model hidden layer [Wang et al. \(2022\)](#), [Zhao and Bilen \(2023\)](#), [Zhao et al. \(2023\)](#) or training dynamics [Cazenavette et al. \(2022\)](#), [Cui et al. \(2023\)](#), [Zhao et al. \(2021\)](#), [Zhou et al. \(2022\)](#) with real datasets. Generative DD approaches [Cazenavette et al. \(2023\)](#), [Gu et al. \(2024\)](#), [Su et al. \(2024\)](#), [Wang et al. \(2023\)](#), [Zhang et al. \(2023\)](#) utilizes the prior knowledge in image generative models to synthesize data samples. They are unable to address the adaptability of temporal granularity based on different level of semantic complexities. Meanwhile, some existing DD works utilized teacher models' knowledge [Sun et al. \(2024\)](#), [Yin et al. \(2023\)](#) to supervise the synthesization process and increase the representativeness of the synthetic datasets. However, when trying to optimize the temporal granularity of synthetic data, their approach does

not provide applicable solutions. VDS [Wang et al. \(2024b\)](#) specifically addresses the challenges on video datasets. It aims to naively reduce the information of all videos into a static image and interpolate it to compensate for dynamic information. As a result, it shows a marginal improvement compared to previous image-level DD approaches, leaving the unique challenge of preserving temporal information unaddressed.

2.2. Reinforcement learning for efficient video understanding

We propose to utilize a Q-learning Reinforcement Learning (RL) framework to obtain the optimal temporal partitioning. RL [Sutton \(1988\)](#), [Sutton and Barto \(1990\)](#) has been developed as a machine learning optimization technique. It is able to address problems requiring sequential decision makings and non-differentiable choices with delayed supervision signal [Burnetas and Katehakis \(1997\)](#). Among the RL algorithms, Q-learning algorithm [Watkins \(1989\)](#) is a model-free approach which does not require the modeling of the environment, and has a good mathematical guarantee on optimization convergence [Watkins and Dayan \(1992\)](#). These characteristics of RL algorithms make them well-suited for addressing temporal partitioning problems in videos without incurring excessive computational costs. There are existing works applying RL to address unique challenges for videos by adaptively focusing on informative regions [He et al. \(2019\)](#), [Wang et al. \(2021, 2020\)](#), [Wu et al. \(2019\)](#), none of existing works have attempted to adaptively reduce resource consumption with temporal partitioning.

3. Methodology

The propose method aims to decouple the optimization of temporal partitioning and the optimization of synthetic videos. In our DAViD pipeline, the RL process and the DD process happen iteratively, ensuring stable updates of RL policies and synthetic videos. We applied Q-learning as a effective RL algorithm due to its good math guarantee of convergence [Watkins and Dayan \(1992\)](#).

3.1. Problem definition

Given the original video dataset with M classes and total n data points.

$$V_{\text{real}} = \{(\mathbf{x}_i^r, \mathbf{y}_i^r)\}_{i=1}^n$$

$$\mathbf{x}_j^r \in \mathbb{R}^{L \times H \times W \times C}, \quad \mathbf{y}_j^r \in \{1, \dots, M\}$$

where L is the temporal duration of an individual video, H is the height, W is the width, and C is the number of channels. Dataset Distillation aims to obtain a synthetic dataset

$$V_{\text{syn}} = \{(\mathbf{x}_{c,i}, \mathbf{y}_{c,i}) \mid c = 1, \dots, M, i = 1, \dots, I\},$$

where $\mathbf{x}_{c,i} \in \mathbb{R}^{L \times H \times W \times C}$, $\mathbf{y}_{c,i} = c$, so that there are I synthetic videos per class. I is called the Instance Per Class (IPC). The optimization process then solve

$$V_{\text{syn}} = \underset{V}{\operatorname{argmin}} \mathcal{L}_{\text{DD}}(V_{\text{real}}, V) \quad (1)$$

where $\mathcal{L}_{\text{DD}}(\cdot, \cdot)$ is the dataset-distillation loss and V is any possible synthetic video dataset.

In this work, we propose to drive the optimization process with two objectives: (1) To minimize conventional DD loss and (2) To maximize the RL reward function defined to discover better temporal partitioning policies. We formulate the problem as a joint optimization of the pixels and the temporal partitioning on the synthetic videos. Such objectives can be formulated as below:

$$\hat{\tau} = \underset{\tau}{\operatorname{argmax}} R(V_{\text{syn}}(\tau)) \quad (2)$$

$$V_{\text{syn}}(\tau) = \underset{V}{\operatorname{argmin}} \mathcal{L}_{\text{DD}}(V_{\text{real}}, \text{Partition}(V, \tau)) \quad (3)$$

where τ is any possible temporal partition policy, formally defined in Part 3.3, and $V_{\text{syn}}(\tau)$ is the optimal synthetic video given a specific τ . $\hat{\tau}$ is the optimal temporal partitioning policy with the highest reward. In this work, RL reward function is proposed by us, $R(\cdot) = R_{\text{DAViD}}$, which is formally defined in Part 3.2. $\text{Partition}(\cdot)$ is a deterministic temporal cropping and resizing operation (detailed definition in Part 3.3).

3.2. Teacher-in-the-Loop reward function

The key challenge of formulating a RL reward under DD scenario is the lack of ground truth to measure the effectiveness of a DD process. This is because the test set is not visible to the whole framework.

A well-trained teacher model has adequate knowledge to extract features that represent video semantics. We propose to apply the teacher as an encoder, extracting the feature of the synthetic and the real dataset of the same class. A reward is therefore calculated based on the feature space distance between the synthetic and real video. The formal formulation of the reward function can be expressed as:

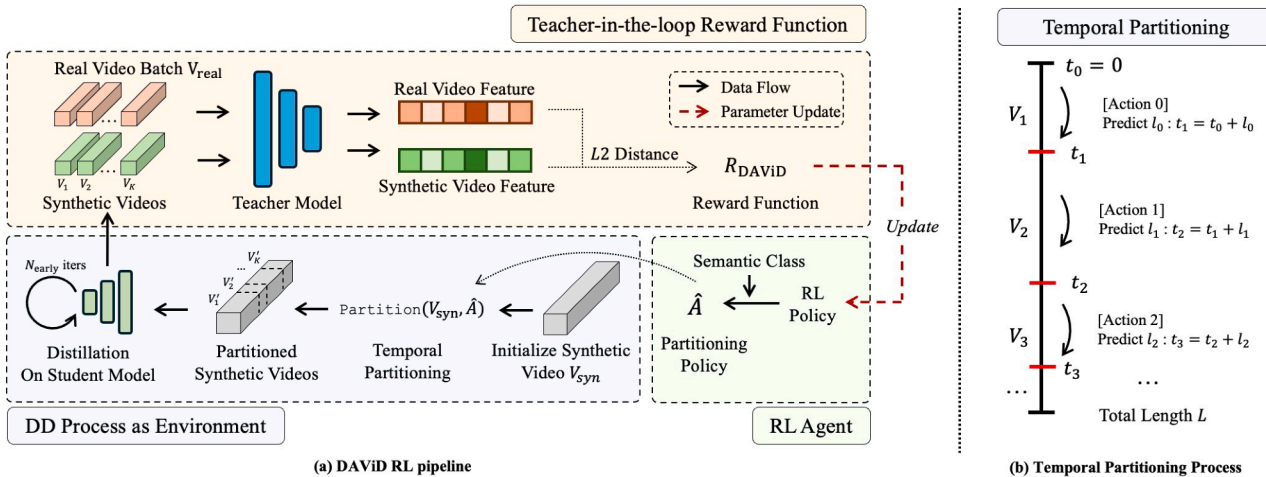


Fig. 4. (a) The distillation process of DAViD utilizes an RL approach to predict the optimal Temporal Partitioning for synthetic data. The learning process of the RL agent is guided by a Teacher-in-the-loop Reward Function, which supervises its policy update. The RL agent predicts the partitioning action for the synthetic videos to adapt to different video semantics. The DD process serves as the environment for the RL agent. We utilize a teacher model to evaluate the feature distance between the synthetic and real data under a given temporal partitioning policy, thereby defining the reward function. (b) The actions of the RL agent are to predict the temporal lengths of next segment, and the actions continues until the full length of the synthetic video is covered by predicted segments.

Algorithm 1 Partition(V, τ).

```

1: Input: Input Video Set:  $V$ 
2: Output: Partitioned Video Set:  $V_{out}$ 
3: Utils: Standard Interleave Repeating Function:
   interleave_repeat( $\cdot, \cdot$ )
4:
5: Temporal Location  $P = 0$ 
6:  $V_{out} = \{\}$ 
7:
8: for  $c$  in range  $[0, M]$  do
9:   for  $t$  in range  $[0, T_c]$  do
10:    Length  $l_t = A_{c,t}$ 
11:     $\mathbf{x}_c = \{x_i \mid y_i = c\}_{i=0}^{|V|}$  and  $x_i \in \mathbb{R}^{L \times H \times W \times C}$ 
12:    for  $x$  in  $\mathbf{x}_c$  do
13:       $x' = x^{P:l_t, \dots, \dots}$  and  $x' \in \mathbb{R}^{l_t \times H \times W \times C}$ 
14:       $\bar{x} = \text{interleave\_repeat}(x', L)$  and  $\bar{x} \in \mathbb{R}^{L \times H \times W \times C}$ 
15:       $V_{out}.append((\bar{x}, c))$ 
16:    end for
17:     $P \leftarrow P + l_t$ 
18:  end for
19: end for
20:
21: Return  $V_{out}$ 

```

$$R_{\text{DAViD}}(V_{\text{syn}}(\tau)) = \frac{1}{1 + \|\Phi_{\text{teacher}}(V_{\text{syn}}(\tau)) - \Phi_{\text{teacher}}(V_{\text{real}})\|_2} \quad (4)$$

Here Φ_{teacher} is the teacher model trained on the original full dataset.

3.3. RL framework for dataset distillation

As illustrated by Fig. 4, we propose to formulate the problem of finding an optimal temporal partitioning into a RL problem. Under this formulation, distillation process on the student model is the environment of the RL agent, generating rewards for different temporal partitioning.

An action taken by the RL agent is any possible of numbers of frames to be partitioned into next segment, which can be formulated as:

$$\text{Action Space } \mathcal{A} = \{1, 2, \dots, L\} \text{ and } A_{c,t} \in \mathcal{A} \quad (5)$$

where L is the targeted total number of frames in the video, t is the index of action step, and c is the class index since the actions are conditioned on different video classes. When $\sum_0^t A_{c,t} \geq L$, the temporal partitioning process ends for class c .

The state of the RL process is the frame index of the previous temporal segment's ending. The state of the RL process can be formulated as the following:

$$\text{State Space } S = \{0, 1, \dots, L\} \text{ and } S_{c,t} \in S \quad (6)$$

We propose to update the RL agent policy guided by a Teacher-in-the-Loop reward function, which will be introduced in part 3.2. Detailed optimization process is illustrated in Algo.2. We propose to use Q-learning as the RL framework for DAViD. On one hand, it is challenging to apply model-based RL approaches and to model the DD process as an environment. This is because of the randomness of the outcome of the DD process and the computational requirement of a DD optimization to converge. On the other hand, we expect the RL algorithm to be easy to converge to the optimal policy. As a model-free learning process with good theoretical guarantee Watkins and Dayan (1992), Q-learning is a suitable RL algorithm for this task.

The RL agent takes actions conditioned on semantic classes. Therefore a semantic-conditioned Q table can be defined as: $Q = \{Q_c\}_{c=0}^M$.

Therefore the action taking process could be formulated as:

$$A_{c,t} = \begin{cases} \underset{A}{\operatorname{argmax}} Q_c(S_{c,t}, A), & \text{if } \epsilon < p \\ \operatorname{rand_choice}(\mathcal{A}), & \text{if } \epsilon \geq p \end{cases} \quad (7)$$

where $t = 0, 1, \dots, T_c$. $A \in \mathcal{A}$ is any possible action in the action space. T_c is the total partitioning step for a semantic class, and $c = 0, 1, \dots, M$.

The Q-table, actions and states of the RL agent is conditioned on each semantic class, allowing Adaptive Partitioning policies. Here ϵ is a random variable of a uniform distribution in the interval of $[0, 1]$, and $\operatorname{rand_choice}(\cdot)$ is a random selection with equal probabilities for each option. Via this mechanism, the RL agent is able to exploit the performance of the estimated best temporal partitioning while maintains a probability of $(1 - p)$ for each action taken to explore new choices.

The Q-values update can be described as following:

$$Q_c(S_{c,t}, A_{c,t}) = Q_c(S_{c,t}, A_{c,t}) + \alpha [R_{t+1} + \gamma \max_A Q_c(S_{c,t+1}, A) - Q_c(S_{c,t}, A_{c,t})] \quad (8)$$

Q_c is conditioned on semantic classes, and all classes share one same reward function R_{t+1} . Thus, temporal partitioning policies across different classes can work cohesively to produce a more effective synthetic video dataset overall. For our DAViD approach, reward signal is sparse due to the nature of DD task, and is received when the temporal partitioning ends and distillation is conducted on the student. Namely, $R_t = 0$ for $0 \leq t \leq T_c$, and $R_{T_c+1} = R_{\text{DAViD}}(V_{\text{syn}}, V_{\text{real}})$, γ is the discount factor for the future rewards, and α is the learning rate of the Q values.

After the temporal partitioning ends, a sequence of actions is obtained for each class of synthetic videos:

$$\tau = \{\{A_{c,t}\}_{t=0}^{T_c}\}_{c=0}^M \quad (9)$$

We initialize the synthetic videos with noise, and conduct Temporal Partitioning on them given τ . The detail of this process is shown in Algorithm 1. `interleave_repeat( $x, L$ )` in Algorithm 1 is a standard interleave-repeating along the temporal dimension of input x to reshape it to the target temporal duration L ³.

3.4. Efficient temporal policy learning

On the basis of RL algorithm's high sampling efficiency from temporal partitioning policy space, we further accelerate the temporal policy learning process by accelerating the distillation on student model. The inner loop distillation process on the student takes N iterations to update V_{syn} , which is significantly larger than the outer loop RL iterations Iter_{RL} , limiting the overall efficiency for temporal policy learning.

To achieve more efficient inner loop process, we apply Distribution Matching (DM) loss Zhao and Bilen (2023) to the inner loop dataset distillation loss $\mathcal{L}_{DD}$ on student model, which does not require updating of student parameters. Meanwhile, since the relative advantage of different partitioning policies becomes apparent at the early stage of the DD process, we set N_{early} as a limitation on the total iterations for the inner loop.

$$N_{\text{early}} = \operatorname{int}(\beta N), \beta \in (0, 1] \quad (10)$$

Here β is an hyper-parameter and $\operatorname{int}(\cdot)$ is rounding to an nearest integer. Such design could largely improve the efficiency of searching, while still being able to generate meaningful reward signal to the RL agent. After the partitioning policies of the RL agent converges, we optimize $V_{\text{syn}}(\tau)$ based on the optimal policy for a total of N iterations.

Table 1

Evaluation on TR-UCF, HMDB51 and UCF101. The Relative Size reflects the ratio of size between the original and the distilled dataset. Here "KeyFrame" is simple optical-flow based implementation and "KeyFrame[†]" is a recent VLM based Key Frame Selection technique [Dirfaux \(2025\)](#). We propose Temporal-Redundant UCF(TR-UCF) as a new subset of UCF101 dataset, where more dynamic classes with various level of temporal redundancies are collected in this subset. Our approach achieved significantly higher performance advantage on TR-UCF, while obtaining improvement by a large margin on HMDB51 and UCF101 as well. The experimental results indicate the importance of dynamic-awareness during video DD.

Dataset	TR-UCF			HMDB51			UCF101		
IPC	1	5	10	1	5	10	1	5	10
Relative Size	1.1%	5.3%	10.6%	1.0%	4.9%	9.8%	1.1%	5.3%	10.6%
Random	6.38 ± 0.5	18.3 ± 0.1	22.1 ± 0.4	4.6 ± 0.5	6.6 ± 0.7	8.3 ± 0.4	3.5 ± 0.1	5.1 ± 0.5	7.9 ± 0.3
KeyFrame	6.34 ± 0.2	19.0 ± 0.4	-	5.1 ± 0.5	7.1 ± 0.3	-	4.1 ± 0.2	6.1 ± 0.3	-
KeyFrame [†]	9.7 ± 1.0	18.0 ± 0.7	24.2 ± 0.9	3.9 ± 0.7	7.8 ± 0.5	9.5 ± 0.7	5.7 ± 0.4	11.0 ± 0.3	14.7 ± 0.4
DM	12.7 ± 0.4	20.1 ± 0.1	30.2 ± 0.4	6.1 ± 0.2	8.0 ± 0.2	14.4 ± 0.3	7.5 ± 0.4	13.7 ± 0.1	19.4 ± 0.5
MTT	17.1 ± 0.2	27.9 ± 0.1	-	6.6 ± 0.5	8.4 ± 0.6	-	11.4 ± 0.7	18.7 ± 0.1	-
VDS	12.2 ± 1.0	23.6 ± 0.8	32.6 ± 0.2	8.6 ± 0.5	10.3 ± 0.6	15.2 ± 0.7	7.8 ± 0.0	14.6 ± 0.2	18.3 ± 0.3
DAViD (Ours)	26.6 ± 0.8	38.4 ± 0.6	44.0 ± 0.7	12.0 ± 0.3	20.9 ± 1.1	27.8 ± 0.6	15.9 ± 0.3	23.6 ± 0.3	27.4 ± 0.7
Full Dataset	45.97 ± 0.5			28.59 ± 0.69			25.83 ± 0.0		

Algorithm 2 $\text{DAViD}(V_{\text{real}}) \rightarrow V_{\text{syn}}, \tau$.

```

1: Input: Training Set:  $V_{\text{real}}$ 
2: Output: Synthetic Video Set:  $V_{\text{syn}}, \tau$ 
3: Initialize: Total episodes of RL:  $\text{Iter}_{\text{RL}}$ ; Early-stage iterations of DD:  $N_{\text{early}}$ ; Total iterations of DD:  $N$ ; Q-table:  $Q$ ; Synthetic video set:  $V_{\text{syn}}$ 
4: function TEMPORAL_POLICY_LEARNING
5:   for  $i$  in  $\text{range}(\text{Iter}_{\text{RL}})$  do
6:     Compute  $\tau$  using Eq. (9)
7:     for  $n$  in  $N_{\text{early}}$  do
8:       Update  $V_{\text{syn}}$  using Eq. (3)
9:     end for
10:    Update  $\tau$  using Eq. (2)
11:   end for
12:   Return:  $Q$ 
13: end function
14: function Synthesize( $Q$ )
15:   Compute  $\tau$  using Eq. (9)
16:   for  $n$  in  $\text{range}(N)$  do
17:     Update  $V_{\text{syn}}$  using Eq. (3)
18:   end for
19:   Return:  $V_{\text{syn}}, \tau$ 
20: end function

```

4. Experiments

4.1. Resource-limited setting for edge device

We conduct the experiments under limited resources to verify the effectiveness of our method for edge device scenario. We conduct distillation with our DAViD approach for IPC = 1, 5, 10. Meaning only 1, 5 or 10 synthetic videos are produced for each class of the dataset. Under such setting, the synthetic dataset has only a relative size of around 10% of the original full dataset at IPC = 10. Furthermore, we aim to achieve our results by training on light-weight models (around 3.6M parameters) to simulate the setting of limited computational resource from a model's perspective.

4.2. Datasets

To evaluate the effectiveness of our approach, we conduct video DD experiment on the most classic and commonly used Video Recognition datasets. The evaluation datasets include UCF101 [Soomro et al. \(2012\)](#), HMDB51 [Kuehne et al. \(2011\)](#), Something-SomethingV2 (SSv2)

[Goyal et al. \(2017\)](#) and Kenetics400 (K400) [Carreira and Zisserman \(2017\)](#).

For an edge device scenario, these datasets are extremely computationally heavy from both time and memory perspective. UCF101 dataset focuses on human actions, covering semantic meanings from sports to daily-life activities. With 101 different classes, it has a total of 13,320 videos, and has 131 videos per class on average and average time duration from 2 to 14 seconds per class. Meanwhile, HMDB51 is the smallest dataset among these, with 6849 videos in total and 51 classes. The semantic meanings of HMDB51 covers a different set of classes compared to UCF101. Meanwhile, we also extend the evaluation to more challenging datasets. Specifically, SSv2 has 220,847 videos in total with 174 classes, and K400 consists of 400 classes and in total 306,245 videos samples.

4.2.1. Temporal-Redundant UCF (TR-UCF)

We propose a new split of the UCF101 dataset to study the effect of various temporal redundancy. We categorize the classes based on their level of redundancy, selecting 50 classes with more dynamic semantic meanings, where addressing the temporal redundancy problem is particularly challenging.

To identify videos with more dynamic semantic meanings, we first train the target model on the full dataset. Then, we retrain the model on a static version of the dataset. To produce the static version of the dataset, we randomly select and retain only one frame from each video sample, namely $V_{\text{static_real}}$. This modification eliminates the temporal information in the dataset. We then calculate the difference of performance between training on the full dataset and training on the static version. Such calculation could be formulated as:

$$\Delta = \text{eval}(\text{train}(\Phi_{\text{student}}, V_{\text{real}})) - \text{eval}(\text{train}(\Phi_{\text{student}}, V_{\text{static_real}})) \quad (11)$$

Here Φ_{student} is the randomly initialized student model. $\text{train}(\cdot, \cdot)$ is the training process that returns the trained model, and $\text{eval}(\cdot)$ is the evaluation process that returns the class-wise performance of a given model on the V_{real} test set. Therefore higher Δ value means more dynamic and complex semantics of the class.

For video datasets, we typically expect a significant drop in training performance when temporal information is removed, as $V_{\text{static_real}}$ lacks the time-dependent features that are crucial for most video recognition tasks. However, we observed Δ is near zero or even has negative values for some classes. We identify these classes as those that do not rely heavily on temporal information for recognition. We rank the classes in the original UCF101 dataset based on the Δ values, and use the classes with the highest 50 Δ values to form a new subset of UCF101: Temporal-Redundant UCF (TR-UCF).

³ Please see detailed implementation of interleave-repeating at https://docs.pytorch.org/docs/stable/generated/torch.repeat_interleave.html

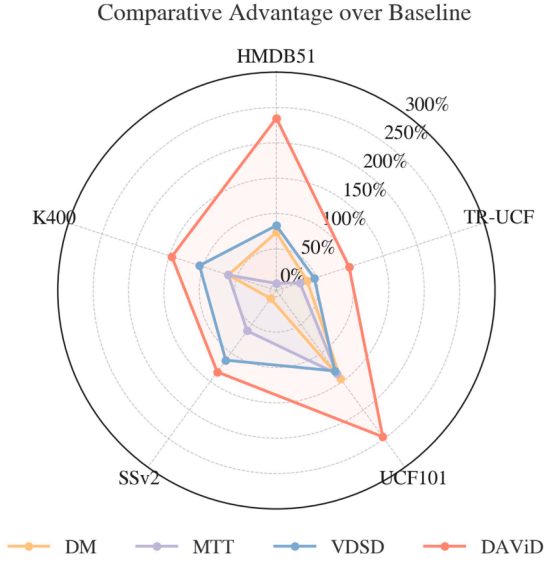


Fig. 5. We evaluated performance improvement on multiple video recognition benchmarks, HMDB51, TR-UCF, UCF101, SSv2 and K400. The y-axis of this radar chart shows the comparative advantage (in percentage) over the baseline of random selection method. We obtained consistent performance improvement compared to existing approaches including DM Zhao and Bilen (2023), MTT Cazenavette et al. (2022) and VDSD Wang et al. (2024b).

Thus, the TR-UCF framework serves as a robust dataset to evaluate the effectiveness of video DD, particularly in distinguishing the influence of temporal dynamics on model performance.

4.3. Implementation details

We apply Distribution Matching Zhao and Bilen (2023) approach as the image-level distillation approach in DAViD. The student model is a 3-layer C3D Tran et al. (2015) following the setting of VDSD Wang et al. (2024b). For a fair comparison, we follow the same data loading shape as VDSD to ensure experimental consistency. A 112×112 spatial size is applied for UCF101, HMDB51 and TR-UCF, and a 64×64 size for SSv2 and K400.

We used SGD optimizer with a learning rate of 10 and a momentum of 0.95 for the synthetic videos. The optimization process was conducted on three NVIDIA RTX 3090 GPUs for 5×10^3 iterations. For the training of the student model on the synthetic dataset, we trained the model for 1×10^3 iterations with a learning rate of 0.01 and applied early stop technique. The early-stage iteration percentage β of DD process in the Temporal Policy learning process is set to be 0.02. We apply a diminishing learning rate for Q-learning, which follows inverse-time decay schedule, $\alpha = \frac{1}{t}$. The discount factor for future reward γ is set to be 0.5.

4.4. Evaluation metrics

We utilized standard evaluation metrics commonly employed in Dataset Distillation to assess the effectiveness of our synthetic dataset. Specifically, for a fixed Instances Per Class (IPC) setting, we evaluated the performance of the synthetic dataset by measuring the accuracy of the student model on the original test set. Temporal partitioning is applied to the evaluation process as we proposed in DAViD. For each evaluation, we train the student model on the synthetic dataset, and test their performance on the original test set. We repeated such experiment process for 10 times to obtain the mean accuracy and standard deviation in the tables.

4.5. Comparison with existing methods

As illustrated in Table 1 and Fig. 2, our approach has achieved improvement by a large margin over existing methods, and achieves nearly lossless distillation. As IPC increases, performance for DAViD increases fast, and reached the performance of the original dataset at IPC = 10, while existing approaches has shown a slow increase and plateaued early. Especially, we observed our approach preserved a strong performance on TR-UCF, which is a significantly more challenging subset. This is because TR-UCF, as mentioned in part 4.2.1, contains various level of temporal redundancies and requires strong dynamic awareness. In this comparison, our approach has shown strong capability to reserve temporal information.

As shown in Fig. 5, we further extend the evaluation to more benchmarks. The performances on HMDB51, TR-UCF and UCF101 are evaluated under IPC = 10. The ones on K400 and SSv2 are evaluated under IPC = 5 due to the heavy computational cost. By comparing across benchmarks, we observed that our approach yields large improvements compared to baselines and previous works.

It is worth mentioning that our approach uses Distribution Matching as the backbone, while VDSD's performance shown in this table uses MTT as the backbone pipeline, which requires significantly more computational resources. As illustrated in Fig. 6, our approach achieves significantly larger accuracy gain by consuming relatively light-weighted computational resources.

4.5.1. Class-wise analysis

As illustrated in Fig. 7, we grouped different classes in UCF101 dataset according to their Δ values, and plot performances under IPC = 1. Group 0 has the smallest Δ , meaning the classes require the least temporal information to be recognized, and vice versa for Group 10. The value of each bar is the average performance among the classes in a certain group. The performance degradation of existing approach is significant from Group 5 to Group 10. In contrast, our approach shows a more balanced result among classes, while achieving a universal improvement across different classes.

Furthermore, given different level of dynamic semantics, the difficulty of classes varies significantly. We can come to a conclusion that existing approaches not only largely ignored the problem of various temporal redundancy, but also struggles to deal with hard samples in the dataset. This largely limits the effectiveness and robustness of the DD process. This indicates that the inter-class balance problem requires more attention in the task of video DD.

4.6. Ablation studies

In this ablation study, we aim to verify the efficiency of our Temporal Policy Learning mechanism, and the necessity to adapt to different video semantics when aiming to reduce temporal redundancies.

4.6.1. Temporal policy learning efficiency

As illustrated in Table 2, we compared with two baseline settings, Grid Search and naive RL Search. Grid Search means to iterate through the whole Temporal Partitioning Policy action space $\mathcal{A}$ with a predefined step length. Naive RL search means the RL agent will wait until the DD process to converge before getting the reward for an action. The RL Search with Early Stage DD is the approach proposed in our DAViD.

We show the total computational workload in TFLOPs to evaluate the efficiency of different temporal policy learning process. The results shows that our proposed Efficient Temporal Policy Learning approach is 2 orders of magnitude more efficient compared to Grid Search regarding GPU Time.

4.6.2. Comparison with other optimization frameworks

As illustrated in Table 3, RL has shown a clear advantage as an optimization frame to discover temporal partitioning policies. In this

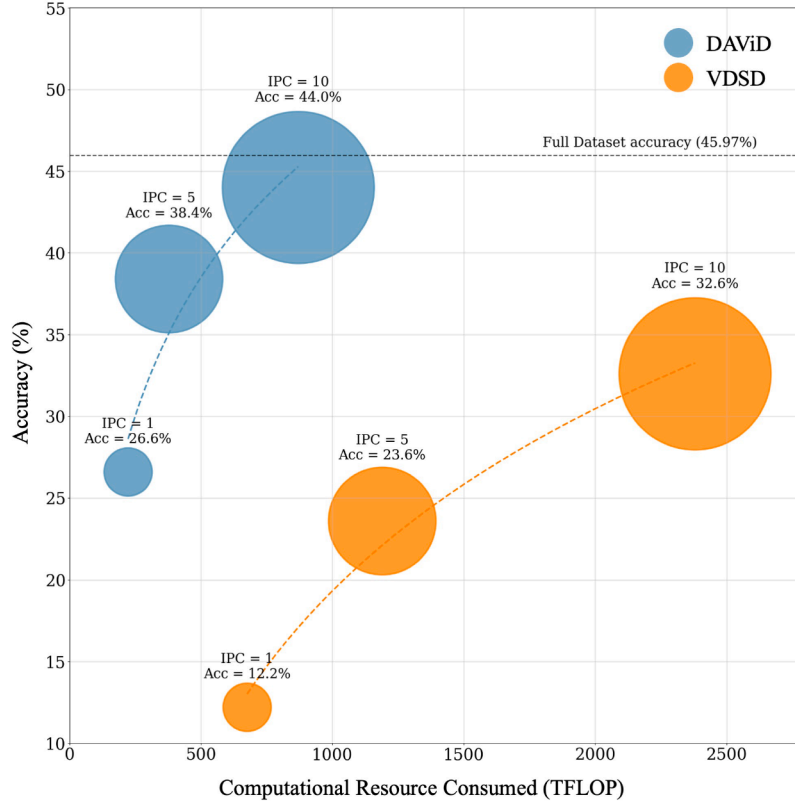


Fig. 6. Computational cost comparison with the previous SOTA approach VDSO Wang et al. (2024b). Cost-efficiency is evaluated by plotting accuracies against computational resources consumed. The area of the scatter point is proportional to the synthetic dataset size. As shown in the figure, DAVID not only achieved a faster accuracy improvement as computational resources consumed increases, but also constantly outperform VDSO from both accuracy and cost-efficiency perspective under the same IPC.

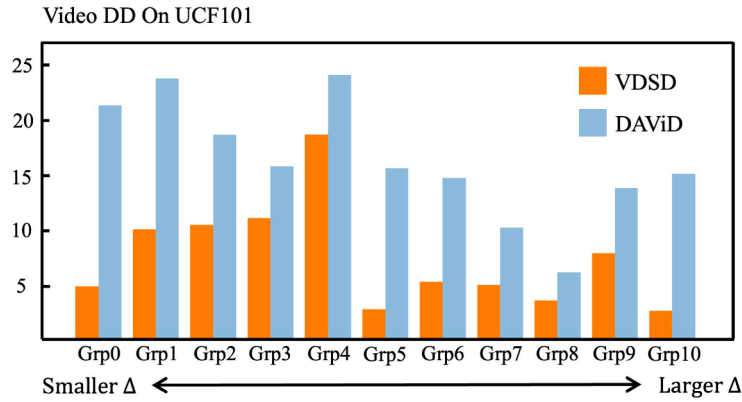


Fig. 7. Distillation effectiveness affected by dynamic semantic meanings. As illustrated in this figure, we divided UCF101 dataset into 11 groups according to their Δ values. From left to right, Group 0 is the most static group of videos and Group 10 is the most dynamic group of videos. Existing approaches only performs relatively well on Group 1–4, and their performance dramatically drops on other groups. In contrast, our proposed method not only dramatically outperforms existing approaches, but also have a balanced performance across all different semantic meanings.

experiment, we give the same computational budget to Bayesian Optimization (BO), Genetic Algorithm (GA) and RL, meaning they have to same total number of temporal partitioning policies that can be explored. Their performance for video recognition accuracy is evaluated on TR-UCF dataset.

We attribute the advantage of RL to its sequential decision-making formulation and low complexity of search space. Given any state, the search space of RL is of size $L - 1$, namely all possible lengths of the next segment. However, for BO and GA, the problem cannot be formulated into a sequential decision-making problem, and all potential ac-

tions taken on individual frames are combinatorial. In these cases, the size of their search space is 2^L , because an action on any frame could lead to a different partitioning policy. Therefore, given the same limited computational budget, BO and GA fail to find a temporal partitioning policy as effective as that of the RL framework.

4.6.3. Video semantics adaptiveness

In this part of ablation study, we aim to verify the effectiveness of the RL guided video DD pipeline. As illustrated in Fig. 4, we tested out three different cases. Case A is conducting distillation without the tem-

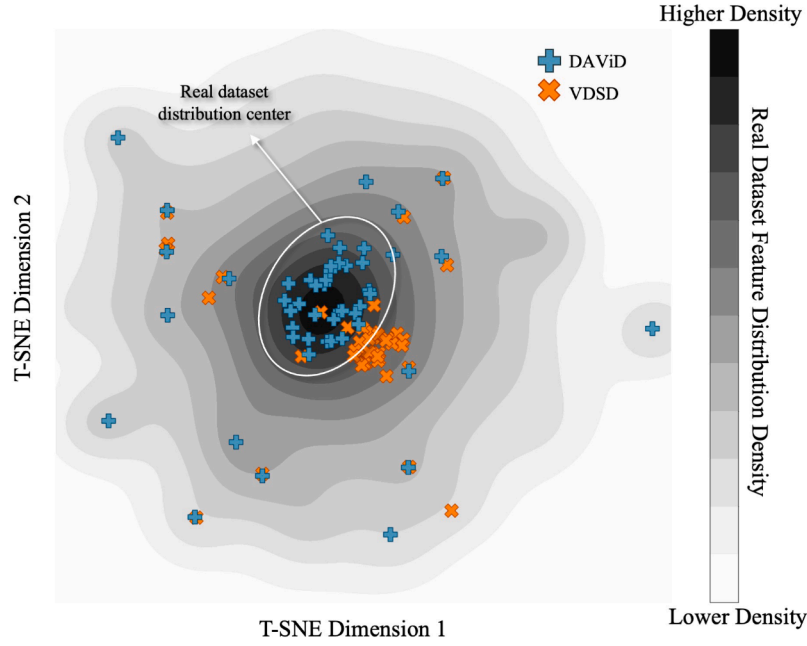


Fig. 8. Feature space T-SNE visualization of synthetic videos on TR-UCF dataset. The gray-scale background shows the density of real dataset distributions. Our approach not only has a better representation of majority classes closer to the center of real data feature distribution, but also shows a better diversity representing minority classes on the edge. This indicates that dynamic-awareness is crucial to the representativeness of synthetic video datasets.

Table 2

Comparison of total computational workload in TFLOPs for different Temporal Policy learning methods on HMDB51. We aim to verify the contribution of RL Search and the early-stage DD mechanism to the overall temporal policy learning efficiency. The approach we proposed is 2 orders of magnitude more efficient compared to Grid Search.

Temporal Policy Learning Setting	TFLOPs	
	IPC = 1	IPC = 5
Grid Search	3417.6	5546.5
RL Search	1715.9	2894.3
RL Search w. Early Stage DD	32.0	56.9

Table 3

Comparison with more optimization frameworks. We apply Bayesian Optimization (BO) and Genetic Algorithm (GA) to conduct adaptive temporal partitioning and distillation on TR-UCF dataset. Given the same computational budget, these two frameworks perform significantly worse than RL. This is contributed by the large search space of both BO and GA, where each frame has to be considered as potential partitioning location.

Optimization Framework	Accuracy	
	IPC = 1	IPC = 5
Bayesian Optimization	18.5 $\pm$ 0.7	23.4 $\pm$ 0.3
Genetic Algorithm	17.9 $\pm$ 0.5	24.8 $\pm$ 0.6
RL	26.6 $\pm$ 0.8	38.4 $\pm$ 0.6

poral policy guidance of RL, and naturally this case does not allow a learnable temporal partitioning policy. Case B is using a universal temporal policy for all video classes. As shown in this ablation experiment, such universal temporal partitioning across classes is not able to adapt different semantic meanings of the video, and it caused an sub-optimal performance. The third case is the full implementation of our DAViD approach.

Table 4

Adaptiveness to video semantics. Case A utilizes neither RL-guided temporal partitioning nor Temporal Policies adaptive to video semantics, this case is equivalent to a naive image-level dataset distillation approach. Case B with only temporal partitioning does not have a dynamic-awareness of video semantics.

	RL Guided Partitioning	Adaptive to Semantics	TR-UCF Accuracy
A			15.4 $\pm$ 0.2
B	✓		20.0 $\pm$ 0.5
DAViD	✓	✓	26.6 $\pm$ 0.8

The results of the ablation indicates the importance of adaptable temporal partitioning for different video semantics. Such observation verified the problem we observed in Fig. 1, and it shows the necessity of dynamic-awareness when conducting video dataset redundancy reduction.

4.7. Visualizations

As illustrated in Fig. 8, we visualized the 2D T-SNE similarity of the features among the real dataset videos, the synthetic video produced by VDS Wang et al. (2024b) and our DAViD. The gray-scale background shows the density of real dataset feature distributions, and each point in the figure is the mean feature of a class of the synthetic data.

On one hand, it is observed that our approach clearly represents the majority classes in the center better, while VDS Wang et al. (2024b) produces synthetic videos with a feature distance gap from the real dataset. Such gap in feature distribution could be caused by the loss of temporal information. If most feature preserved by VDS is the static features, the feature could still have certain similarity to the real dataset, but unable to represent the dynamic feature from the real videos.

On the other hand, for the classes far from the majority, our approach returns a clearly more diverse and representative feature distribution compared to VDS. The feature visualization clearly shows the importance and effectiveness of preserving temporal information and dynamic-awareness.

4.8. Limitations

Despite the large margin of improvement, this work is still largely constrained by the resource scale of training and data storage. There could be more exploration on heavy-weight datasets and student models. Different teacher-student architectures can be further verified in the future to explore the generalizability and robustness of the technique. Furthermore, future explorations could be conducted to extend such dynamic aware distillation methods to more edge-device related tasks and modalities such as video generation Zhan et al. (2024), multi-view computer vision Li et al. (2024), digital twins Chen et al. (2024a,b), Yang et al. (2024) and vehicular networks Li et al. (2025). We believe the above limitations could inspire more works contributing to efficient on-device training research.

5. Conclusion

This is the first work to address the problem of Adaptive Temporal Partitioning for video Dataset Distillation (DD) tasks. We propose a Dynamic-Aware Video Distillation (DAViD) with a RL framework to adaptively predict the optimal temporal partitioning policy for different video semantics. An efficient temporal policy learning mechanism is proposed, and two orders of magnitudes acceleration is achieved on the search for optimal temporal partitioning policies compared to grid search baseline. Our approach achieved SOTA accuracy on multiple video DD benchmarks under a setting of limited dataset and model sizes. With such large margin of performance improvement, our dynamic-aware DD approach brings potential to address current limitation of video DD research on more heavy-weight models and on the generalization to more architectures and modalities.

CRedit authorship contribution statement

Yinjie Zhao: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Heng Zhao:** Writing – review & editing, Supervision; **Yew-Soon Ong:** Supervision; **Bihan Wen:** Supervision; **Joey Tianyi Zhou:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work is supported by the National Research Foundation, Singapore under its Digital Trust Centre Innovation Grant (DTC Award No: DTC-IGC-02).

References

- Ben-Yakov, A., & Henson, R. N. (2018). The hippocampal film editor: Sensitivity and specificity to event boundaries in continuous experience. *The Journal of Neuroscience*, 38(47), 10057–10068. <https://doi.org/10.1523/JNEUROSCI.0524-18.2018>
- Bumetas, A. N., & Katehakis, M. N. (1997). Optimal adaptive policies for markov decision processes. In *Mathematics of operation research*.
- Cai, H., Gan, C., Zhu, L., & Han, S. (2020). TinyTL: Reduce memory, not parameters for efficient on-device learning. In *NeurIPS*.
- Carreira, J., & Zisserman, A. (2017). Quo vadis and action recognition? a new model and the kinetics dataset. In *CVPR*.
- Cazenavette, G., Wang, T., Torralba, A. A. E., & Zhu, J.-Y. (2022). Dataset distillation by matching training trajectories. In *CVPR*.
- Cazenavette, G., Wang, T., Torralba, A., Efros, A. A., & Zhu, J.-Y. (2023). Generalizing dataset distillation via deep generative prior. In *CVPR*.
- Chen, J., Yi, C., Du, H., Niyato, D., Kang, J., Cai, J., & Shen, X. (2024a). A revolution of personalized healthcare: Enabling human digital twin with mobile AIGC. *IEEE Network*, 38(6), 234–242. <https://doi.org/10.1109/MNET.2024.3366560>

- Chen, J., Yi, C., Okegbile, S. D., Cai, J., & Shen, X. (2024b). Networking architecture and key supporting technologies for human digital twin in personalized healthcare: A comprehensive survey. *IEEE Communications Surveys & Tutorials*. <https://doi.org/10.1109/COMST.2023.3308717>
- Cui, J., Wang, R., Si, S., & Hsieh C.-J. (2023). Scaling up dataset distillation to imagenet-1k with constant memory. In *ICML*.
- Dirfaux, F. (2025). Focus: Efficient keyframe selection for long video understanding. In *arXiv:2510.27280*.
- Gim, I., & Ko, J. (2022). Memory-efficient DNN training on mobile devices. In *Mobisys*.
- Goyal, R., Kahou, S. E., ski, V. M., nska, J. M., Westphal, S., Kim, H., Haenel, V., Fruend, I., Yianilos, P., Mueller-Freitag, M., Hoppe, F., Thureau, C., Bax, I., & Memisevic, R. (2017). The “something something” video database for learning and evaluating visual common sense. In *ICCV*.
- Gu, J., Vahidian, S., Kungurtsev, V., Wang, H., Jiang, W., You, Y., & Chen, Y. (2024). Efficient dataset distillation via minimax diffusion. In *CVPR*.
- He, D., Zhao, X., Huang, J., Li, F., Liu, X., & Wen, S. (2019). Read, watch, and move: Reinforcement learning for temporally grounding natural language descriptions in videos. In *AAAI*.
- Huang, K., Yang, B., & Gao, W. (2023). Elastictrainer: Speeding up on-device training with runtime elastic tensor selection. In *Mobisys*.
- Jouini, O., Sethom, K., Namoun, A., Aljohani, N., Alanazi, M. H., & Alanazi, M. N. (2024). A survey of machine learning in edge computing: techniques, frameworks, applications, issues, and research directions. *Technologies*. <https://doi.org/10.3390/technologies12060081>
- KHAN, M. A., BACCOUR, E., CHKIRBENE, Z., ERBAD, A., HAMILA, R., HAMDI, M., & GABBOUJ, M. (2016). A survey on mobile edge computing for video streaming: Opportunities and challenges. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2022.3220694>
- Khemlani, S. S., Harrison, A. M., & Trafton, J. G. (2015). Episodes, events, and models. *Frontiers in Human Neuroscience*. <https://doi.org/10.3389/fnhum.2015.00590>
- Kim, B., Lee, J.-T., Yang, S., & Chang, S. (2023). Scalable weight reparametrization for efficient transfer learning. In *ICASSP*.
- Kuehne, H., Jhuang, H., Garrote, E., Poggio, T., & SerreT. (2011). Hmdb: A large video database for human motion recognition. In *ICCV*.
- Kurby, C. A., & Zacks, J. M. (2008). Segmentation in the perception and memory of events. *Trends in Cognitive Sciences*, 12(2), 72–79. <https://doi.org/10.1016/j.tics.2007.11.004>
- Lee, S., Weerakoon, M., Choi, J., Zhang, M., Wang, D., & Jeon, M. (2022). Carn: Hierarchical episodic memory for continual learning. In *DAC*.
- Li, B., Zheng, C., Zhu, W., Mai, J., Zhang, B., & Ghanem, P. W. B. (2024). Vivid-ZOO: Multi-view video generation with diffusion model. In *NeurIPS*.
- Li, X., Deng, R., Wei, J., Wu, X., Chen, J., Yi, C., Cai, J., Niyato, D., & Shen, X. (2025). AIGC-driven real-time interactive 4-d traffic scene generation in vehicular networks. *IEEE Network*, 39(6), 261–269. <https://doi.org/10.1109/MNET.2025.3545609>
- Lin, J., Chen, W.-M., Lin, Y., Cohn, J., Gan, C., & Han, S. (2020). Mccnet: Tiny deep learning on iot devices. In *NeurIPS*.
- Lin, J., Zhu, L., Chen, W.-M., Wang, W.-C., Gan, C., & Han, S. (2022). On-device training under 256KB memory. In *NeurIPS*.
- Liu, X., Li, B., Chen, S., & Xu, Z. (2025). A survey on multi-user privacy issues in edge intelligence: State of the art, challenges, and future directions. *Electronics*. <https://doi.org/10.3390/electronics14122401>
- Lopez-Paz, D., & Ranzato, M. (2017). Gradient episodic memory for continual learning. In *NeurIPS*.
- Lopez-Paz, D., & Ranzato, M. (2019). Deep learning with edge computing: A review. *Proceedings of the IEEE*, 107(8), 1655–1674. <https://doi.org/10.1109/JPROC.2019.2921977>
- Nguyen, L.-T., Quélennec, A., Nguyen, V.-T., & Tartaglione, E. (2025). Beyond low-rank decomposition: A shortcut approach for efficient on-device learning. In *ICML*.
- Paissan, F., Nadalini, D., Rusci, M., Ancilotto, A., Conti, F., Benini, L., & Farella, E. (2024). Structured sparse back-propagation for lightweight on-device continual learning on microcontroller units. In *CVPR Workshops*.
- Patil, S. G., Jain, P., Dutta, P., Stoica, I., & Gonzalez, J. E. (2022). Poet: Training neural networks on tiny devices with integrated rematerialization and paging. In *ICML*.
- Radvansky, G. A., & Zacks, J. M. (2014). Event Cognition. Oxford University Press.
- Sargent, J. Q., Zacks, J. M., Hambrick, D. Z., Zacks, R. T., Kurby, C. A., Bailey, H. R., Eisenberg, M. L., & Beck, T. M. (2013). Event segmentation ability uniquely predicts event memory. *Cognition*, 129(2), 241–255. <https://doi.org/10.1016/j.cognition.2013.07.002>
- Singh, R., & Gill, S. S. (2023). Edge AI: A survey. *Internet of Things and Cyber-Physical Systems*, 3, 71–92. <https://doi.org/10.1016/j.iotcps.2023.02.004>
- Soomro, K., Zamir, A. R., & ShahM. (2012). Ucf101: A dataset of 101 human actions classes from videos in the wild. In *CoRR*.
- Su, D., Hou, J., Gao, W., Tian, Y., & Tang, B. (2024). D4m: Dataset distillation via disentangled diffusion model. In *CVPR*.
- Sun, P., Shi, B., Yu, D., & Lin, T. (2024). On the diversity and realism of distilled dataset: an efficient dataset distillation paradigm. In *CVPR*.
- Sutton, R. S. (1988). Learning to predict by the methods of temporal differences. In *Machine-mediated learning*.
- Sutton, R. S., & Barto, A. G. (1990). Time derivative models of pavlovian reinforcement. Tran, D., Bourdev, L., Fergus, R., Torresani, L., & Paluri, M. (2015). Learning spatiotemporal features with 3d convolutional networks. In *ICCV*.
- Wang, K., Gu, J., Zhou, D., Zhu, Z., Jiang, W., & You, Y. (2023). Dim: Distilling dataset into generative model. In *arXiv:2303.04707*.
- Wang, K., Zhao, B., Peng, X., Zhu, Z., Shuo Yang, S. W., Huang, G., Bilen, H., Wang, X., & YouY. (2022). Cafe: Learning to condense dataset by aligning features. In *CVPR*.
- Wang, T., Zhu, J.-Y., Torralba, A., & EfrosA. A. (2018). Dataset distillation. In *arXiv:1811.10959*.

- Wang, X., Tang, Z., Guo, J., Meng, T., Wang, C., Wang, T., & Jia, W. (2025). Empowering edge intelligence: A comprehensive survey on-device AI models. *ACM Computing Surveys*. <https://doi.org/10.1145/3724420>
- Wang, Y., Chen, Z., Jiang, H., Song, S., Han, Y., & Huang, G. (2021). Adaptive focus for efficient video recognition. In *ICCV*.
- Wang, Y., Dong, M., Shen, J., Wu, Y., Cheng, S., & Pantic, M. (2020). Dynamic face video segmentation via reinforcement learning. In *CVPR*.
- Wang, Z., Wu, F., Yu, F., Zhou, Y., Hu, J., & Min, G. (2024a). Federated continual learning for edge-AI: A comprehensive survey. *arXiv:2411.13740*.
- Wang, Z., Xu, Y., Lu, C., & Li, Y.-L. (2024b). Dancing with still images: Video distillation via static-dynamic disentanglement. In *CVPR*.
- Watkins, C. J. C. H. (1989). Learning from delayed rewards. In *Ph.d. thesis, king's college*.
- Watkins, C. J. C. H., & Dayan, P. (1992). Technical note q-learning. In *Machine learning*, 8, 279–292.
- Wu, Z., Xiong, C., Ma, C.-Y., Socher, R., & Davis, L. S. (2019). Adaframe: Adaptive frame selection for fast video recognition. In *CVPR*.
- Yang, L., Rakin, A. S., & Fan, D. (2022). Rep-net: Efficient on-device learning via feature reprogramming. In *CVPR*.
- Yang, Y., Shi, Y., Yi, C., Cai, J., Kang, J., Niyato, D., & Shen, X. (2024). Dynamic human digital twin deployment at the edge for task execution: A two-timescale accuracy-aware online optimization. *IEEE Transactions on Mobile Computing*, 23(12), 12262–12279. <https://doi.org/10.1109/TMC.2024.3406607>
- Yin, Z., Xing, E., & Shen, Z. (2023). Squeeze and recover and relabel: Dataset condensation at imagenet scale from a new perspective. In *NeurIPS*.
- Zacks, J. M., Braver, T. S., Sheridan, M. A., Donaldson, D. I., Snyder, A. Z., Ollinger, J. M., Buckner, R. L., & Raichle, M. E. (2001). Human brain activity time-locked to perceptual event boundaries. *Nature Neuroscience*, 4, 651–655.
- Zacks, J. M., Speer, N. K., Swallow, K. M., Braver, T. S., & Reynolds, J. R. (2007). Event perception: A mind-brain perspective. *Psychological Bulletin*, 133(2), 273–293. <https://doi.org/10.1037/0033-2909.133.2.273>
- Zhan, Z., Wu, Y., Gong, Y., Meng, Z., Kong, Z., Yang, C., Yuan, G., Zhao, P., Niu, W., & Wang, Y. (2024). Fast and memory-efficient video diffusion using streamlined inference. In *NeurIPS*.
- Zhang, D. J., Wang, H., Xue, C., Yan, R., Zhang, W., Bai, S., & Shou, M. Z. (2023). Dataset condensation via generative model. In *arXiv:2303.04707*.
- Zhang, Y., Gu, T., & Zhang, X. (2020). Mldroidlite: A release-and-inhibit control approach to resource-efficient deep neural networks on mobile devices. In *Sensys*.
- Zhao, B., & Bilen, H. (2023). Dataset condensation with distribution matching. In *WACV*.
- Zhao, B., Mopuri, K. R., & Bilen, H. (2021). Dataset condensation with gradient matching. In *ICLR*.
- Zhao, G., Li, G., Qin, Y., & Yu, Y. (2023). Improved distribution matching for dataset condensation. In *CVPR*.
- Zhou, Y., Nezhadarya, E., & BaJ. (2022). Dataset distillation using neural feature regression. In *NeurIPS*.